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When Persuasion Is Too Persuasive: An Empirical Analysis of Product Returns in Livestream e-Commerce

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Abstract

Livestream e-commerce features live social interactions that enable real-time, face-to-text communications among a streamer and many customers. We posit such live social interactions generate persuasion effect that eventually leads to higher product returns. Using data from a leading livestream e-commerce firm, we compare the product returns between live sessions with rerun sessions where the only difference is the presence of live social interactions in live sessions but not in rerun sessions, and develop econometric models to estimate the effect of live social interaction intensity on product returns. We find that product returns are higher for the live sessions than for the rerun sessions, and that the magnitude of the effect increases with live social interaction intensity, confirming the adverse persuasion effect in livestream e-commerce. We also find that the persuasion effect of live social interactions on product returns is stronger when the streamer has more experience in hosting livestream sessions and the product price is lower. We estimate that one standard deviation increase in live social interaction intensity can increase the product return rates by as much as 11.68 percentage points in terms of absolute product return rates, which is more than 30% of the average product return rates. Robustness checks, theoretical and managerial implications are also discussed.

Keywords

Live social interaction, Livestream e-commerce, Persuasion effect, Expectation confirmation theory, Product returns, Retail operations

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1 Introduction

Livestream e-commerce has shown explosive growth in recent years. In the US, according to Coresight Research (Larson, 2021; Snow, 2021), the livestream e-commerce market is estimated to be worth $6 billion in 2020 and is projected to reach $11 billion and $25 billion in 2021 and 2023, respectively. In China, livestream e-commerce has developed into a larger business sector with market value of $150 billion and $300 billion in 2020 and 2021, respectively (Arora et al., 2021; Larson, 2021; Snow, 2021). A survey shows that almost 30% of the population in China watched livestreams in 2020 (Greenwald, 2020). Attracted by the growth potential of this sector, firms such as Amazon.com, Facebook, TikTok and Nordstrom in the US, Alibaba's Taobao, Douyin and Kuaishou in China and Douglas in Europe have become early adopters that invested in developing the livestream e-commerce business (Larson, 2021). For example, in December 2020, Walmart launched a livestream e-commerce pilot on TikTok, which attracted seven

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times more viewers than expected and increased the number of followers of Walmart's TikTok account by 25% (Perez, 2021).

Livestream e-commerce differs from existing shopping models in several ways. Building on online shopping in e-commerce that leverages web technologies (De et al., 2013), livestream e-commerce creates a virtual venue that is similar to offline stores for retailers to provide prepurchase services to customers (Musalem et al., 2021), such as oral product introductions or try-on shows. In contrast to live chat services in e-commerce that offer private text-to-text communications (Sun et al., 2021; Tan et al., 2019), livestream e-commerce supports public face-to-text communication, enabling one streamer to serve many customers simultaneously. In addition, in contrast to online video advertising (e.g. Kumar and Tan, 2015), livestream e-commerce allows the streamer (i.e. the seller) to interact with their customers in real time.

As a new shopping trend, livestream e-commerce has narrowed the gaps between conventional online shopping and in-store shopping. During the limited time that a livestream e-commerce session lasts, customers swamp into the virtual venue to watch the streamer, interact with the crowd and make purchases, as if they are attending a social entertainment event. For brands, retailers and marketplaces, livestream e-commerce provides a high-touch way to strengthen live-time customer engagement which involves video chats, sales, consultations and virtual services (Larson, 2021). It is estimated that livestream e-commerce may increase sales conversion rate by tenfold as compared to conventional channels, thereby improving brand appeal and differentiation (Arora et al., 2021).

Livestream e-commerce is different from conventional e-commerce primarily because of its live social interaction function that facilitates customers' both shopping and socializing needs. Prior literature has reached a consensus in general that livestream e-commerce is beneficial to firms as livestream e-commerce can increase sales conversion (Chen et al., 2019; Song et al., 2021). For example, using 2-week livestream e-commerce data from Taobao in women's apparel, Song et al. (2021) find that product characteristics, streamer characteristics and consumer engagement are important factors that drive e-commerce live streaming sales. However, higher conversion rates only show one side of the story of live e-commerce. Customers may choose to return their purchased products. If the product return rates are also higher, it would be difficult to conclude whether livestream e-commerce is beneficial. To the best of our knowledge, none of the prior studies have examined the product returns in livestream e-commerce. To fill the research gaps, we examine the product return rates from livestream e-commerce. Our study is among the first attempts to investigate the impacts of live social interactions on product returns.

In particular, we collected transaction-level data from a leading online retailer that is a pioneer in livestream e-commerce and empirically examine the impacts of live social interactions in livestream e-commerce on the propensity of product returns at the order level. Built on the persuasion effect of live social interactions and expectation confirmation theory (ECT), we posit that live social interactions during livestream e-commerce sessions result in higher product return rates, and that the increased product return rates are moderated by the streamer experience, assortment size and product price. We first test the existence of the positive effect of live social interactions in livestream e-commerce on product returns by comparing the product returns in the same livestream and rerun sessions, between which the only difference is whether there are live social interactions, using propensity score matching (PSM). We then utilize econometric models to estimate the effects of live social interactions and the moderating factors on product returns for the purchases made during livestream sessions. The key findings are that the product return rates are higher for the orders placed during live sessions than those for the rerun sessions, and that higher live social interaction intensity leads to more product returns. And, the effects of live social interactions on product returns are strengthened by streamer experience, but weakened by product price. Our research points to an important managerial implication for livestream e-commerce firms that they should incentivize streamers by integrating the sales volume with the return rate into their performance metrics, which would benefit the net profit of the firm.

Our research makes a number of important empirical and theoretical contributions. First, we study the adverse effect of the live social interactions in the livestream e-commerce. Prior research on social interactions and livestreams has offered a wealth of findings on the advantages of these activities, but few works have examined the disadvantages of social interactions in livestream e-commerce in particular, and e-commerce in general. Our research fills this gap by empirically examining the adverse effects of live social interactions on product returns associated with livestream e-commerce. Second, we theoretically demonstrate several underlying mechanisms that affect the adverse persuasion effect in livestream e-commerce. Prior literature usually emphasizes the value of persuasion for immediate decisions, whereas our research sheds light on the downside of persuasion in affecting subsequent decisions after a consumer is persuaded to make an immediate decision rashly.

The remainder of this research is organized as follows. The background of livestream e-Commerce and our empirical context are introduced in Section 2. Section 3 reviews the related literature that covers the research topics of online reviews, synchronous social interaction and livestreaming. The underlying theory and hypotheses are demonstrated in Section 4. Section 5 presents the data and empirical analysis results. We conclude the research in Section 6.

2 Livestream e-Commerce

2.1 Industry Overview

Livestream e-commerce is an emerging business model that bridges the gaps between online retailing and in-store shopping (Larson, 2021). It allows people to create their own

livestream channels, generally hosted on an online retailing platform, to sell products to viewers through interaction and entertainment. A shopper can experience the whole purchase journey without leaving the broadcast, including picking a product from the product list featured in the livestream, browsing the product details, observing live product demonstrations and making a purchase (Hallanan, 2020). More importantly, a shopper can also interact with the streamer through live texting and receiving the streamer's responses in the livestreaming, or interact with other shoppers through live texting. The live interaction imitates the experience of in-store shopping where shoppers may interact with sales associates and other customers. To facilitate this interactive process and subsequent transactions in a live stream, the streamer plays a key role in engaging with shoppers in a high-touch way involving video chats, live selling, consultations and virtual services (Larson, 2021).

In the early phase of livestream e-commerce, streamers are mainly top influencers and celebrities. They play an important role in connecting supply and demand in livestream e-commerce by selecting products with deep or even exclusive discounts from various manufacturers (i.e. the supply side) and attracting customers (i.e. the demand side) by leveraging their enormous social influence. In a stream session that lasts for a few hours, products with individual purchase links and limited inventory are launched one by one, accompanied by exciting music and the streamer's calls to action to purchase the products (Greenwald, 2020).

Along with livestream e-commerce's evolution, the market has shifted from top influencers acting as streamers to microstreamers that include store-run streamers and independent micro influencers, in both the US and China (Greenwald, 2020; Hallanan, 2021). Top influencers are usually key opinion leaders in a particular field; when they act as streaming hosts, they have much greater appeal and influence on consumer behaviors than microstreamers. Compared with the top or independent microinfluencers that collaborate with multiple brands and launch dozens of un-preannounced products in succession within a livestream session, the store-run streamers present products owned by the focal store and have become more and more popular. For example, 90% of the largest brands on Alibaba's Tmall platform have already engaged in store-run livestream e-commerce in 2020; 70% of the livestreaming sessions on Taobao Live platform are store-run (Hallanan, 2021).

2.2 Empirical Context

We collaborated with a mobile commerce firm that specializes in selling high-end products online in China. The firm provides direct and third-party sales to end customers mainly through a mobile application. The product categories include luxury bags, watches, clothing, footwear, accessories, jewelry, cosmetics and skincare, electronic devices, etc. The firm was a top three firm in terms of market share in its segmentation market, generated more than $900 million in revenue, had over 420,000 Stock Keeping Units (SKUs) and facilitated more than 4 million transactions in 2020. Due to the luxury attribute, both brand-new and used items are sold on the platform. To ensure the authentication of items, the firm builds a certification center to provide authoritative appraisal services online and offline.

The firm started the livestream e-commerce in January 2020. On their mobile commerce platform, the livestream category page displays both live and rerun sessions, through which customers can make purchases. Live sessions are listed at the top of the page, followed by rerun sessions. Once a live session ends, its session recording is immediately turned into a rerun session and available in the rerun session section. All the sessions are hosted by store-run streamers, among them 18.5% are from the firm and others are from third-party sellers.

A customer can easily access a session by clicking on the session link. Figure 1(a) shows a screenshot of a sample live session that consists of a streamer, product offerings, a public live comment area, an interactive chat box, a button for free likes or virtual-valued gifting and a share button and so on. Figure 1(b) depicts, in contrast, a screenshot of the corresponding rerun session. The product offerings are located in the bottom left corner of the session page. By clicking the product offerings, customers can view a product list, including product information, price and pictures and then purchase the products in a way similar to typical online shopping without leaving the stream session. The product offerings icon shows the total number of products demonstrated by the streamer in the session; that is, the assortment size. In Figure 1, for example, the assortment size is 213.

Compared to a rerun session, a live session has several unique features that constitute live social interactions but are not available in the rerun session: (a) the video is live in real time; (b) live chat box is available; (c) the numbers of likes and virtual gifts are available and (d) live total number of views is presented. All these features are available in real time and dynamically changing during the entire livestream session. The rerun session does not have the public live comment area and the interactive chat box because the customers are not able to interact with the streamer any more. They are also not presented with the social interactions during the live session. That is, a customer can view the rerun session to obtain exactly the same information content from the streamer but will not be able to interact with her via the chat box and view the social interactions during the live session. In an ongoing live session, the customer can engage in live social interactions with the other customers and ask the streamer questions. Table 1 summarizes the differences between live and rerun sessions.

The messages left by customers in the public comment area are usually of one of several types: (a) asking the streamer to introduce a specified product, try it on, zoom in on the product to observe more details, or provide purchase-related information; (b) communicating with (or responding to) the

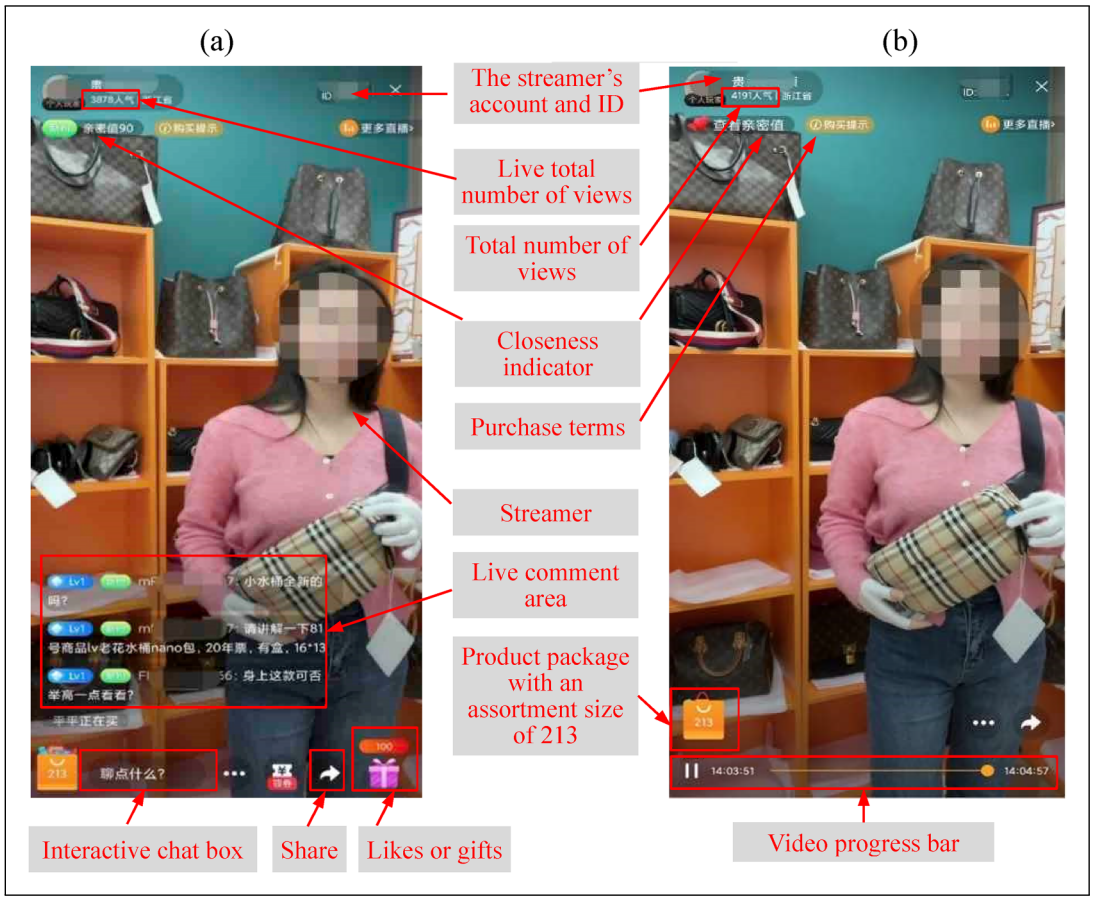


Figure 1. Screenshots of sample live and rerun sessions. (a) Live session. (b) Rerun session.

Table 1. Comparison of live and rerun sessions.

	Live Session	Rerun Session
The streamer's account and ID	Yes	Yes
Number of views	A real-time value	An end-status value of the live session
Closeness indicator	Yes	Yes, and with a click
Purchase terms	Yes	Yes
The streamer	Yes	Yes
Product package	Yes	Yes
Live comment area	Yes	No
Interactive chat box	Yes	No
Button for share	Yes	Yes
Button for likes or gifts	Yes	No
Video progress bar	No	Yes

streamer and other customers or (c) commenting on a product, the streamer, or other customers. The streamer typically responds to the questions and comments in the chat box by talking or showing in livestream video (instead of via texting in the live comment area). Hence, the information delivered in the video is exactly the same for live and rerun sessions. This creates a highly controlled environment for us to test our hypotheses on the persuasion effect resulting from the live social interactions among customers in live sessions.

3 Literature Review

3.1 Online Reviews

Extensive literature has been developed to investigate the impacts of online product reviews on sales (e.g. Kwark et al., 2021; Wang et al., 2022a). Major findings from the literature are that online product reviews have a significant impact on sales elasticity (see Floyd et al., 2014 for a review), and that electronic word of mouth (WOM) has a positive impact on

sales and that the impact of effectiveness differs across platforms, products and metrics (see Babić Rosario et al., 2016 for a review).

The impacts of online reviews on product returns, however, remain largely underexplored, while several studies have examined the topic. Minnema et al. (2016), using the electronics and furniture categories, point out that review volume may not have significant effects on product returns but that an overly positive review valence can increase the product returns in the long-term due to greater expectation disconfirmation in an online retailer context. In a different setting involving a specialty retailer of apparel, accessories and home furniture, Sahoo et al. (2018) find that product returns can be significantly reduced by higher review volume, lower rating dispersion and the presence of more helpful reviews. More recently, for an online apparel retailer website that introduced a new fit reference function (i.e. a feature to evaluate the customer's body size and the objective size of the apparel purchased) in addition to the existing fit valence function (i.e. a feature to assess the subjective size of the apparel purchased), Wang et al. (2021) show that reviews on fit valence alone can reduce product returns only when the rating dispersion is very low.

Almost all studies in this stream of literature are related to the existent post purchase reviews on the product information page, where customers can browse the static information at any time. In such cases, social interaction is minimal, if present at all in the form of textual replies from after-sales service representatives on only some of the reviews (e.g. Gu and Ye, 2014). Our research is different from prior studies by focusing on live social interactions that are available during livestream e-commerce sessions in real time using both text and live streaming.

3.2 Synchronous Social Interaction

Our study is also related to the literature that empirically investigates the consequences of synchronous social interactions. A general theme in findings in this literature is that, in a richer online media context where information, video and comments coexist, users are more willing to participate in online social interaction. For example, Guo et al. (2020) demonstrate that a real-time indicator of the number of platform users within a geospatial neighborhood location can encourage users to contribute more content to a geosegmented, crowdsourced mobile community application, and Jiang et al. (2013) find that individuals may voluntarily disclose private data to obtain more social rewards in synchronous online social interaction when there is greater media richness.

Live chat has been increasingly adopted by online retailers as a channel to interact with customers synchronously and facilitate transactions. Two recent studies use data to empirically analyze the impacts of live chat on customer purchase decisions. Tan et al. (2019) show that live chat can increase the purchase probability of tablets by 15.99%, and this effect is strengthened by the product sales volume but weakened by the seller feedback score. Sun et al. (2021) confirm the positive impacts of live chat on sales conversion and further show that this effect is stronger for products with less comprehensive information or higher perceived value (i.e. with either a higher rating or a lower price).

This stream of literature focuses on the impacts of synchronous social interactions among customers only or between customers and sellers only, whereas the social interactions in our setting involve all parties (sellers and all customers) and occur in both directions. Furthermore, in our study, the synchronous social interactions in the form of customers' live comments and retailer live streaming videos are public, in contrast to the private features of interactions in prior studies.

3.3 Livestreaming

As an emerging business model in industry, livestreaming has attracted the interest of a growing number of scholars. Works in this literature stream can be classified into two categories based on whether the streamer sells products during a livestream session or not. We name the first category content-based livestreaming without sales, during which the streamer broadcasts content and interacts with viewers through video in real time and earns tips or payments from viewers. Given that viewer engagement is one of the primary performance metrics in this category, prior research has identified the underlying motivations (Hilvert-Bruce et al., 2018), and the correlated factors regarding streamer emotion (Lin et al., 2021), streamer characteristics (Zhao et al., 2021), audience size (Lu et al., 2021) and session operations (Zeng et al., 2020).

The second category relates to ecommerce-based livestreaming, which differs from online shopping in a video advertising context (Kumar and Tan, 2015) due to the presence of live social interaction. In addition to customer engagement, studies in this category also consider sales performance due to the e-commerce component (Pan et al., 2022; Wongkitrungrueng and Assarut, 2020; Chen et al., 2023). Research has shown that the livestreaming can significantly increase product sales because of the real-time sales information to streamers (Chen et al., 2019; He et al., 2021) and that the sales performance is associated with product features, streamer characteristics, customer engagement such as the number of real-time comments, shares and likes, traffic acquisition promotion, price discounts and short videos (Cheng et al., 2023; Duan et al., 2023; Song et al., 2021; Zhang et al., 2022).

Our study is related to the second stream of the literature on livestream e-commerce but differs from prior research in this stream. While prior studies have elucidated the benefits of adopting livestream e-commerce technology and identified factors that may boost sales, few studies have considered product returns as a performance metric, though there are extensive studies on product returns in OM society (e.g. De et al., 2013; Gallino and Moreno, 2018; Guide and Van Wassenhove, 2001;

Hwang et al., 2022; Rao et al., 2014; Sahoo et al., 2018; Shang et al., 2017, 2019; Wang et al., 2021, 2022b). Our research fills this gap by studying the product returns resulting from livestream e-commerce, thus, providing a more complete picture of the benefits of livestream e-commerce.

4 Hypothesis Development

4.1 Persuasion Effect

As discussed above, livestream e-commerce is a new form of e-commerce that connects sellers and consumers through livestreaming online. It employs advanced technology to facilitate public, real-time, face-to-text interactions among all participants—one streamer and many customers—at a specific time in an online-only space. A unique feature of livestream e-commerce is that it enables live social interactions that play a role in informing and persuading consumers.

Prior research has shown that both the information and persuasion effects play roles in influencing customers in the prepurchase stage albeit in distinct ways (Ching and Ishihara, 2012; Sun et al., 2021). Studies in general agree that information provision directly informs consumers about product-specific attributes and that persuasion induces customers to make a purchase, although the interpretation of the persuasion effect differs across research settings. For example, in the live chat context, persuasion can emphasize the value of products (Sun et al., 2021), whereas, in the prescription drug context, persuasion can boost the ability of a promotion to create "artificial" product differentiation or brand-specific qualities (Ching and Ishihara, 2012). In some more general contexts, peripheral cues such as source expertise (Karmarkar and Tormala, 2010; Wilson and Sherrell, 1993), source credibility (Jain and Posavac, 2001; Tybout, 1978), the attractiveness or celebrity status of the product endorsers (Petty and Cacioppo, 1986), puffery (Chakraborty and Harbaugh, 2014), or the vividness of information presentation (Herr et al., 1991) are believed to drive the persuasion effect.

In our setting, the information effect refers to the effect of the streamer's ability to present product-related information, including the oral introduction of the product attributes, zoom-ins on different parts of a product, or on-site try-ons or tasting. The persuasion effect refers to the effect of the streamer's efforts to convince consumers to make purchase decisions through peripheral approaches beyond providing objective product information, such as fostering a lively and interactive atmosphere that attracts customer engagement, advertising the merits of products, or setting up professional lighting and studio scenes.

We conjecture that both the information effect and the persuasion effect lead to higher purchase probability. However, different from the information effect that lowers product returns by reducing product uncertainty regarding how well the product meets consumers' needs (Sahoo et al., 2018), the persuasion effect may result in higher product returns. Previous studies have found that the persuasion effect that relies on peripheral cues is less persistent over time and less resistant to changes than the effects driven by product-related cues (Haugtvedt et al., 1994). Thus, if a customer is persuaded to rashly purchase a product, they may change their mind later on and eventually return the product, given that there exists a time lag between online order placement and order receipt. When the persuasion effect is stronger, the customers who bought the products should form a higher expectation of the product performance and thus, are more likely to be disappointed when they receive the actual products. According to the ECT (Oliver, 1980), the discrepancy between the expected performance and actual performance incites feelings of dissatisfaction in customers and eventually leads to product returns (Minnema et al., 2016; Rao et al., 2014; Sahoo et al., 2018).

Recall that our research setting covers livestream sessions and rerun sessions. The key difference between livestream and rerun sessions is that livestream sessions support live social interactions whereas rerun sessions do not. Rerun sessions are recordings of livestream sessions with the same video footages hosted by the same streamer. Hence, the information effects are the same between the pairs of the livestream and rerun sessions, but the persuasion effects are stronger in livestream sessions than rerun sessions because the live social interactions enable a real-time lively atmosphere. Prior research has shown that communication presented in a more vivid manner is more persuasive than that presented less vividly (Herr et al., 1991). The live streaming plus a real-time lively atmosphere may seem more vivid to consumers than product pictures and videos alone, thus generating a stronger persuasion effect. This creates a highly controlled environment for us to test our hypotheses on the persuasion effect resulting from the live social interactions among customers in livestreaming e-commerce. Thus, we hypothesize that livestream e-commerce is associated with a higher product return due to live social interactions. Hence, we propose our first hypothesis:

H1: *The orders placed in livestream e-commerce sessions with live social interactions are more likely to be returned than those placed in the sessions without live social interactions.*

H1 examines the existence of the positive effect of live social interactions on product returns. In the following hypotheses, we hypothesize on the intensity of persuasion effect in the livestream sessions. In particular, we posit the effect in H1 increases in the intensity of live social interactions. That is, the product returns increase with the intensity of live social interactions. More intense live social interactions make customers feel more vivid and are part of the advertising repetition strategy, both of which have been shown to result in stronger persuasion power (Haugtvedt et al., 1994, Herr et al., 1991). In our case, we measure the intensity of live social interaction by the number of live comments.

When the live social interactions become more intense (i.e. more live comments), customers form higher expectations at the prepurchase stage due to stronger persuasion effect, but this does not affect customers' actual experience of product performance in the post-purchase stage. As a result, the larger

gaps between the expected and actual performance will aggravate customer dissatisfaction and generates a higher return tendency. Therefore, we have the following hypothesis.

H2: *The orders placed in livestream e-commerce sessions with higher live social interaction intensity are more likely to be returned.*

4.2 Moderating Effect of Streamer Experience

The persuasion effect posited in H2 may be moderated by streamer experience. Previous literature has studied the positive effect of source expertise on persuasion in various contexts (Karmarkar and Tormala, 2010; Klucharev et al., 2008; Wilson and Sherrell, 1993). For example, through a meta-analysis of source effects in communication and persuasion, Wilson and Sherrell (1993) reveal that source expertise has a greater effect on persuasion than non-expertise related factors such as physical attractiveness or similarity. Similarly, Karmarkar and Tormala (2010) find that source expertise can promote persuasion particularly when uncertainty arguments are expressed. Klucharev et al. (2008) further support the positive effect of source expertise on persuasion with evidence from functional magnetic resonance imaging. Considering that experience is one crucial foundation of expertise, more experienced people may also know better how to leverage persuasion power (Reuber, 1997; Wilson and Sherrell, 1993).

In our setting, during a livestream session that typically lasts for a few hours, the streamer not only publicly responds to customers who contribute in the live comment area but also speaks to the whole audience in real time. As livestreaming involves new ways of communication via technology, the streamer learns by doing on how to read the customers' sentiments in the comments, how to present information and answers effectively, how to accurately address customers' concerns and how to promote sales conversion, thereby improving their communication skills in this new context. Thus, from the view of audiences in livestream e-commerce, when faced with the same level of live social interaction, a more experienced streamer should appear as owning higher source expertise and thus exhibiting stronger persuasion power. Specifically, holding the intensity of live social interaction constant, when the streamer gains more experience in hosting stream sessions, the streamer becomes more persuasive in pushing sales, resulting in larger perception gaps between the expected and actual performance for customers. The larger gaps lead to a greater number of product returns. Hence, we propose the following hypothesis that the streamer experience positively moderates the persuasion effect of live social interaction:

H3: *Greater streamer experience strengthens the effect of live social interactions in livestream e-commerce on product returns.*

4.3 Moderating Effect of Assortment Size

The persuasion effect posited in H2 may also be moderated by a livestream session's assortment size; that is, the number of products recommended by the streamer for sale in the session. Online retailers have long paid close attention to decisions on product assortment problems that optimize the economy of scales and minimize cannibalization among products (Feldman et al., 2022; Li et al., 2023; Peng et al., 2023; Talluri and van Ryzin, 2004; Wang et al., 2023). In livestream e-commerce, the assortment size is a more important decision for the seller because the livestream session has time constraints. The streamer needs to prepare and present all products for the upcoming livestream session to customers over the duration of a session, which usually lasts for a few hours. Because of time constraints, the assortment size is negatively associated with the streamer credibility because of the necessary preparation time. That is, for a given time constraint, when the assortment size is small, a streamer can spend more time learning about these products so that the streamer is more familiar with the products and thus appears more credible in streaming.

Previous literature has found that source credibility impacts the persuasion effect (Jain and Posavac, 2001; Tybout, 1978). For example, in a personal selling context, Tybout (1978) shows that a high-credibility source generates a significantly stronger effect than a low-credibility source on behavioral compliance under a straight persuasion strategy. Jain and Posavac (2001) find that source credibility can increase the persuasion power of a claim promoting a product experience attribute. In our case, a better-prepared streamer better understands the selling points of a product, appears more credible, and thus performs a stronger persuasion effect when interacting with customers during a session. As a result, a streamer who recommends fewer products (i.e. a smaller assortment size) in a session may be able to exert a stronger persuasion effect through live social interactions, leading to higher product returns eventually, than those who recommend more products. Thus, we have the following hypothesis.

H4: *Larger assortment size weakens the effect of live social interactions in livestream e-commerce on product returns.*

4.4 Moderating Effect of Product Price

Product price may also moderate the persuasion effect posited in H2. Prior studies have shown that consumers are less likely to be persuaded by peripheral cues to purchase high-involvement products than low-involvement products, especially with respect to value-relevant involvement (e.g. Johnson and Eagly, 1989). A high-involvement product usually has a high price and requires extensive exploration before finalizing the purchase decision; in this case, it is more difficult for the seller to persuade customers to make a purchase by highlighting the product benefits or leveraging other peripheral cues. In the live chat context, Sun et al. (2021) show that a retailer with a higher price triggers a weaker positive effect

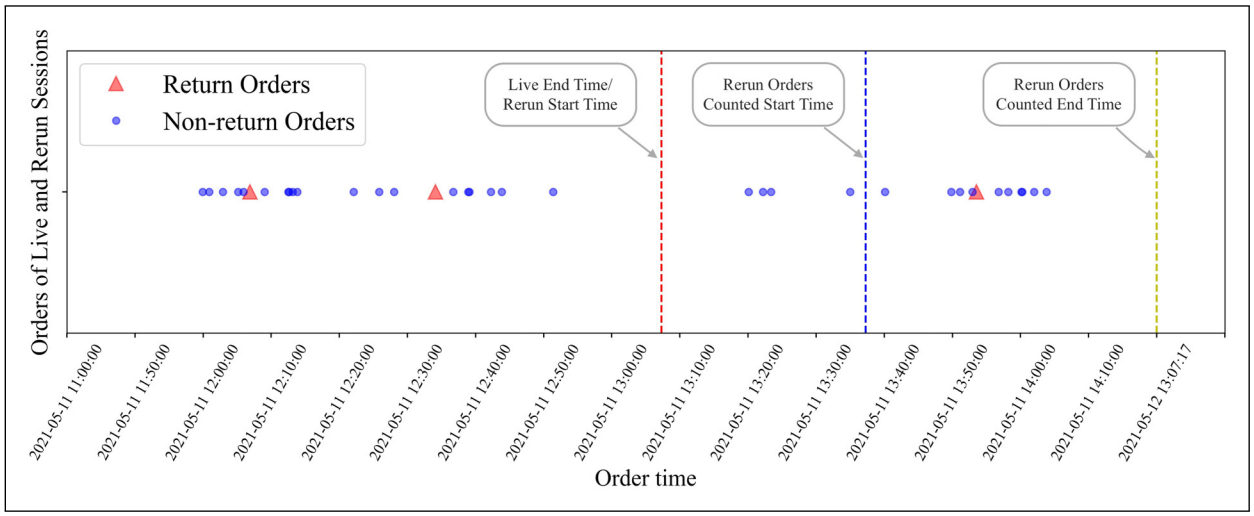


Figure 2. Order timeline in sample live and rerun sessions.

of live chat effort on sales conversion than a retailer with a lower price, supporting that a high price weakens the persuasion effect on sales. Moreover, using a ComScore dataset of Web site visitation and transaction activity, Huang et al. (2009) find that consumers that buy expensive products involve higher breadth (number of pages) and greater depth (time per page) of search, leading to a longer total search time than inexpensive products. At the same time, a customer that spends longer search time is more likely to find a better product that meets her/his preferences, as a consequence, less likely to return the product that has been reviewed carefully in the pre-purchase stage (Ibragimov, 2022). Similarly, in the context of livestream e-commerce, a customer that wants to buy a more expensive product will spend more search time and be more conscious in the search process, and thus less likely to be persuaded by live social interactions in live sessions to rush for an impulse purchase due to the streamer highlighting the product benefits or leveraging other peripheral cues. Therefore, in line with the findings of prior literature, we hypothesize a negative moderating effect of the product price on the persuasion effect of live social interactions on product returns. This leads to the following hypothesis.

H5: *Higher product price weakens the effect of live social interactions in livestream e-commerce on product returns.*

5 Empirical Analysis

5.1 Data

We collected all livestream e-commerce-related data from the firm from the date the firm started the livestream e-commerce business in January 2020 to June 2021. The firm is a public company listed on the US Nasdaq stock market. The data include both session-level data and order-level data. The session-level data include 10,298 sessions from 755 streamers and consist of information such as the number of live comments, likes and time duration of the live session. The order-level data include all orders placed through the livestream e-commerce and consist of information such as order value, order size and whether an order is returned. There are 22,364 orders in total across 25 product categories.

We processed the raw data following two steps. First, to avoid the impact of the COVID-19 pandemic, we kept only transactions placed after May 1, 2020 when COVID-19 started to be under control with fewer than 10 cases in China (World Health Organization, 2020). This step removed 1110 orders. Second, to alleviate outlier issues, we removed transactions above the 99th or below the 1st percentile values for the order value and number of comments (Kwark et al., 2021; Lin et al., 2021). This step removed 630 orders.

As discussed above, the rerun sessions are available for customers to view immediately after the live session is ended. Some customers may have started placing orders during the live session but did not complete the entire purchase process until shortly after a live session ends. To ensure a reliable and accurate categorization of live and rerun orders, we excluded those orders placed within the first 30 min of the rerun session.[1] This step removed 586 orders. Furthermore, we removed orders placed longer than one day after the start time of the rerun session, as these orders are distinct from the normal orders placed during the live and rerun sessions. This step removed 76 rerun orders. As a result, 19,962 orders remain in the final sample, among which 19,748 orders are from livestream sessions and 214 orders are from rerun sessions. Figure 2 presents a timeline for a sample live and rerun session with order placement and returned.

The return policy and shipping cost policy of the firm are uniform across all stores and product categories on the platform. Specifically, customers can return the order freely within seven days of receiving the parcel, as long as the product returned has the same condition as the original and can be

resold. It is stipulated that customers pay for the return shipping costs whatever the reason for returns, and that customers must select a logistics carrier who is on a name list designated by the platform, wherein all logistics carriers on the list provide high-quality delivery services at similar freight rates. Thus, there is no variation on the return and shipping cost policies for customers.

5.2 Product Returns Between Live and Rerun Sessions

To test H1, we compare the product returns associated with the orders from live sessions with those associated with the orders from rerun sessions. As discussed above, it is a highly controlled environment because all conditions are the same except for the availability of live social interactions between the live and rerun sessions. The challenge of comparing the product returns associated with live and rerun sessions is that consumers are not randomly assigned to the live or rerun sessions, although we do not have any reasons to believe they are systematically different. To address this potential issue, we employ PSM, a commonly used method in this situation (Chen et al., 2019; Guo et al., 2020; Wang et al., 2021). PSM pairs a live order with a matched rerun order based on the characteristics of consumers, orders, streamers and sessions. We use the following rules to match the orders:

(1) The orders should be from the same live and rerun session pairs. With the same live and rerun session pair, it controls for the differences in sessions and streamers, such as streamer's styles and product assortments.
(2) The orders should have similar order characteristics, such as similar price, product category and order size, to control for order level heterogeneity.
(3) The orders should have similar consumer characteristics, such as consumer purchase experience, accumulative consumption amount and geographic location, to control for consumer-level heterogeneity.

Specifically, we use a two-step matching procedure to implement the matching process. First, we apply a logit model to obtain the probability of an order placed in the rerun session. The logit equation can be specified as follows:

$$\begin{aligned}\text{Log}\left(\frac{\Pr(Rerun_i=1)}{\Pr(Rerun_i=0)}\right) &= \alpha_0+\alpha_1 CustomerExp_i+\alpha_2 LnCustomerSpend_i \\ &\quad +\alpha_3 OrderValue_i+\alpha_4 OrderSize_i \\ &\quad +\alpha_5 ShippingProvince_i+\alpha_6 Category_i+\varepsilon_i, \end{aligned} \quad (1)$$

where $\Pr(Rerun_i = 1)$ is the propensity of an order would be placed in the rerun session. *CustomerExp* is the customer's purchase experience (Gao et al., 2023), defined as the number of sessions that a customer placed orders before the focal order; *LnCustomerSpend* is the natural logarithm of the cumulative monetary spending of a customer before the focal live session; *OrderValue* is the monetary value of an order; *OrderSize* is the number of items in an order.[2] In addition, we add the shipping province and product category dummies to control for differences in consumers' geographic locations and product categories. The estimation results of this model are presented in Supplemental Table S1. Second, we use propensity scores generated from the first step and match each rerun order with a live order that has the closest propensity in the same session, thereby establishing an order pair. The matching process yields 212 order pairs.

To evaluate the validity of our matching procedure, we conduct a balance test by comparing the differences in variables before and after the matching (Guo et al., 2020; Wang et al., 2021). The results of the balance test are shown in Supplemental Table S2. The results show that the orders for live and rerun sessions are significantly different in terms of three out of four variables before matching but are insignificant for all four variables after matching, indicating successful matching of the order pairs.

With the matched order pairs, we use the McNemar Test to examine whether there are significant differences in return rates between rerun orders and live orders. The McNemar Test has been widely used in empirical research with paired nominal data (e.g. Swan and Combs, 1976). The variable of interest is the binary variable *OrderReturn*, where $OrderReturn = 1$ if the order is returned and $OrderReturn = 0$ otherwise. The null hypothesis is that rerun orders and live orders have equal return rates. Table 2 shows the test results.[3] Panel A shows that, among 212 order pairs, both rerun and live orders are returned for 38 pairs, neither of the orders is returned for 81 pairs, rerun orders are returned but live orders are not returned for 28 pairs, and the rerun orders are not returned but the live orders are returned for 65 pairs. The estimated return rates, presented in Panel B, are 0.49 and 0.31 for live orders and rerun orders, respectively. The test result shows that the difference in return rates is statistically significant ($p = 0.0001$), indicating that the null hypothesis is rejected and that live orders have higher return rates. This result lends support to H1.

5.3 Econometric Models

To test H2–H5 on live social interaction effect on product returns and the three moderating effects, we use live orders only and exclude rerun orders because live social interaction effect is only available in live sessions. The sample for analysis has 19,748 live orders.

The dependent variable is whether an order is returned or not (*OrderReturn*), which is a binary variable that equals 1 if the order is returned and 0 otherwise, as defined above. There are four independent variables. The first independent variable is the natural logarithm of the number of comments in a live session (*LnComment*), which is used to measure the intensity of live social interaction during the session.[4]

Table 2. Paired test of return probability between rerun and live orders.

Panel A: Data layout for matched pairs				
		Control Group (Live Orders)		Total
		Return	Nonreturn	
Treatment group (Rerun orders)	Return	38	28	66
	Nonreturn	65	81	146
	Total	103	109	212
Panel B: McNemar Test				
	Treatment (Rerun Orders)	Control (Live Orders)	Differences	*p* value
	0.31	0.49	−0.17	0.0001

As mentioned in Section 2.2, customers interact with the streamer and the other customers through an interactive chat box, and all interactive messages appear in the public comment area. Hence, a higher number of live comments is associated with a higher intensity of live social interaction. Thus, the number of live comments is a reasonable and valid measure of the live social interaction intensity in H2.

The second independent variable is the streamer's experience (*LnStreamerExp*), which is calculated as the natural logarithm of the total number of live stream sessions hosted by the streamer on the platform before the focal session. The more live stream sessions that a streamer has hosted, the more experienced the streamer is. The third independent variable is the product assortment size for a live session (*AssortSize*); that is, the number of products available to consumers to buy. The fourth independent variable is the natural logarithm of the total monetary value of the order (*LnOrderValue*). *LnStreamerExp*, *AssortSize* and *LnOrderValue* are used to generate the interaction terms with *Comment* to test the moderating effects in H3, H4 and H5.

We add a number of control variables at the session, order, streamer and time levels. First, at the session level, we include the natural logarithm of the number of likes received from viewers during a live session (*LnLikeNum*), the natural logarithm of the total number of views from customers during a live session (*LnViewNum*), the number of customers who placed live orders in a live session (*CustomerNum*) and the time duration of a live session in hours (*LiveDuration*) to control for the differences in sessions. Second, at the order level, we add the order sizes (*OrderSize*), percentage of sales discount (*DiscountRatio*) and order authentication (*IsPlatform*, equals to one if the order is placed in the live session hosted by the official store of the platform and zero otherwise by the third-party retailers) to control for the differences in orders (Shang et al., 2019). Third, at the streamer level, we add the number of fans of the streamer in thousands ($FansNum^5$) and the natural logarithm of the average number of likes received by each streamer per historical live session prior to the focal session (*LnHistStreamerPop*) to control for the streamer's popularity and influence. Fourth, we add the week (*Week*) and the day-of-week (*Weekday*) fixed effects to control for time varying seasonality (Yao et al., 2019). Furthermore, we add the shipping province fixed effect (*ShippingProvince*) and the product category fixed effect (*Category*). Customers located in different provinces may have different socioeconomic statuses, demographics and behavioral patterns and products in different categories may have different return rates (Shang et al., 2017).

Hence, we can specify our models asfollows. Equation (2) is our baseline model without interaction terms to test the impact of live social interaction (H2) and equation (3) adds the interaction terms to test the moderating effects of streamer experience, assortment size and order value (H3–H5).

$$OrderReturn_i = f\begin{pmatrix} LnComment_i, LnStreamerExp_i, AssortSize_i, LnOrderValue_i, LnLikeNum_i, \\ LnViewNum_i, CustomerNum_i, FansNum_i, LiveDuration_i, OrderSize_i, \\ DiscountRatio_i, IsPlatform_i, LnHistStreamerPop_i, Week_i, \\ Weekday_i,\ ShippingProvince_i, Category_i \end{pmatrix} \tag{2}$$

$$OrderReturn_i = f\begin{pmatrix} LnComment_i, LnStreamerExp_i, AssortSize_i, LnOrderValue_i, LnComment_i \\ * LnStreamerExp_i, LnComment_i * AssortSize_i, LnComment_i * LnOrderValue_i, \\ LnLikeNum_i, LnViewNum_i, CustomerNum_i, FansNum_i, LiveDuration_i, OrderSize_i, \\ DiscountRatio_i, IsPlatform_i, LnHistStreamerPop_i, Week_i, Weekday_i, \\ ShippingProvince_i, Category_i \end{pmatrix} \tag{3}$$

Table 3. Variable definitions and descriptive statistics (N = 19,319).

Variable	Mean	Std. Dev.	Min.	Max.	Definition
OrderReturn	0.35	0.48	0	1	Indicator variable for whether an order is returned
Comment	940.61	861.56	1	5080	Number of comments in a livestream session
LastComment	704.47	778.06	0	12,623	Number of live comments in the last live session hosted by the focal streamer
StreamerExp	35.28	32.67	1	213	Number of livestream sessions that a streamer has hosted prior to the focal session
AssortSize	88.22	51.19	1	329	Number of products offered in a livestream session
OrderValue	4227.39	6529.78	140	50,000	Total monetary value for an order
LikeNum	2505.00	5432.24	0	92,446	Number of likes for a livestream session
ViewNum	1769.90	2475.19	3	24,515	Number of views for a livestream session
LiveDuration	5.44	2.19	0.06	24.93	Time duration in hours for a livestream session
CustomerNum	98.01	128.12	1	602	Number of customers who made a purchase in a livestream session
FansNum	8.14	5.17	0.01	21.86	Number of followers for a streamer, in thousands, as of June 8, 2021
OrderSize	1.05	0.31	1	8	Number of items in an order
DiscountRatio	0.02	0.07	0	0.87	Discount percentage for an order
HistStreamerPop	3616.33	4603.85	0	57,618.67	Average number of likes received by each streamer per historical session prior to the focal session
IsPlatform	0.33	0.47	0	1	Indicator variable for whether an order is placed at the live session hosted by the official store of the platform

Note. The smaller sample sizes in this table (19,319) compared with the total sample size (19,748) are due to the null values of historical statistics for the first session. All variables are calculated based on the order level.

where i denotes the order, and the function f may respond to the formulations of probit or linear model used in the following analysis.

Table 3 reports the definitions and summary statistics of the variables and Supplemental Table S4 reports the correlation matrix of variables. To check potential multicollinearity, we compute the variance inflation factor (VIF) scores for all independent and control variables in our model. The VIF scores for all independent and control variables are below 3, which is lower than the commonly accepted cutoff of 10 (Kennedy, 2008), indicating that multicollinearity is not a concern.

5.4 Identification Strategy

Endogeneity may exist in our setting because of unobservable factors. For example, if a customer has a high intention to place an order in a specific live session for some unobserved reasons, this may lead to an increase in the number of comments in the live session, as that person may be more likely to interact with the streamer. Meanwhile, since that person has a high intention to purchase that product, the resulting return probability may also be low. In such a case, purchase intentions may be correlated with both the live comment number and return probability, thus, causing biased estimation of the effects of live social interaction. We conducted the Durbin-Wu-Hausman test[6] on the potential endogeneity of *LnComment.* The test is significant, suggesting *LnComment* may be endogenous.

To address the potential endogeneity issue, we use the instrumental variable (IV) method, which is a standard and common approach used in the literature (Luo et al., 2020; Miguel et al., 2004). A valid IV is correlated with the endogenous variable (*LnComment*) but uncorrelated with the dependent variable (*OrderReturn*). In our case, we use the natural logarithm of the number of live comments in the last live session hosted by the focal streamer (*LnLastComment*) as the IV. *LnLastComment* should be positively correlated with *LnComment* because of the same streamer who has the same communication and persuasion skills. However, *LnLastComment* should not affect the return behaviors of customers in the focal live session because they occur at different times and spaces. Similar instrumental variables have also been widely adopted in the literature (e.g. Tan and Netessine, 2019; Xu et al., 2021).

In our baseline model without interaction terms (equation (2)), we employ the instrumental variable *LnLastComment* to address the endogeneity issue, while in the model with interaction terms (equation (3)), we use the *LnLastComment* to interact with the three moderators to create three more instrumental variables (Angrist and Pischke, 2009: 132). Supplemental Table S5 shows the first-stage results of the IV estimation. As expected, each of the instrumental variables

is significantly and positively correlated with the corresponding endogenous variable. The Cragg-Donald Wald F statistic (497.57) is statistically higher than the Stock-Yogo critical value (16.38), rejecting the null hypothesis that the instrument variable is weak. The under-identification test using the Kleibergen–Paap rk LM statistic (e.g. Luo et al., 2020) shows that the statistic is highly statistically significant, rejecting the null hypothesis that the equation is under identified.

5.5 Estimation Results

Table 4 summarizes the 2nd stage instrumental variable estimation results. Model 1 and Model 2 use 2SLS regressions for the analysis. Since our dependent variable is a binary variable, we also use Probit model with IVs in Model 3 and Model 4 to test our hypotheses. We include all control variables and the fixed effects for the time, shipping province and product category in all models. The estimated coefficients for the independent variable of interest (*LnComment*) and interaction terms are consistent in all four models, demonstrating the robustness of our results. As the coefficients of linear based models are easily to be interpreted (Ho et al., 2022), we discuss the results in Model 2 only.

The coefficient of *LnComment* is positive and significant ($\beta = 0.2074,\ p < 0.001$), indicating that customers are more likely to return the products when the number of comments in the live session is higher. The result suggests that the persuasion effect of more intense live social interaction may induce customers to form a higher pre-purchase expectation at the time of buying that does not match the customers' individual experience of product performance after receiving the product; that is, the discrepancy between a higher pre-purchase expectation and lower real experience. The discrepancy may lead to dissatisfaction, followed by a higher return tendency. Hence, H2 is supported.

The coefficient of the interaction term between *LnComment* and *LnStreamerExp* ($\beta = 0.0404,\ p < 0.001$) is positive and significant, suggesting that if a streamer has more experience hosting livestream sessions, they may possess stronger persuasion power than less experienced streamers, which generates a larger subsequent discrepancy between expectation and experience for customers and results in a higher return probability. Hence H3 is supported.

The coefficient of the interaction term between *AssortSize* and *LnComment* is significant and negative ($\beta = -0.0003$, $p < 0.05$), suggesting that, compared to the case with a larger assortment size, when fewer products are recommended in a live session, the streamer may have more time to prepare and learn each product before the session begins. More preparation work may make the streamer seem more credible during the session, which strengthens the persuasion effect of live social interaction and increases product returns in the end. Therefore, H4 is supported.[7]

The coefficient of the interaction term between *LnOrderValue* and *LnComment* ($\beta = -0.0170,\ p < 0.01$) is significant and negative. In contrast with a higher price order, an order with a lower price usually requires lower involvement; thus, customers are more easily persuaded, leading to higher order return rates. Hence, H5 is supported. Our result that the price attenuates the persuasion effect is consistent with the findings in Sun et al. (2021) for the live chat setting.

Some control variables are significant with expected signs. For example, the coefficients of *LnLikeNum*, *LnViewNum* and *FansNum* are negative and significant in all models, suggesting that higher numbers of likes, views and fans are associated with lower product return rates. The significantly negative coefficients of *IsPlatform* show that orders buying from the official store of the platform have lower product return rates than that from the third-party retailers, indicating that a higher authentication level is beneficial to reduce product returns. The F statistics are significant in all estimations, rejecting the null hypothesis that the coefficients are jointly zero.

5.6 Robustness Tests

We conduct several robustness checks. First, we use a new variable to measure the streamer experience, that is, the natural logarithm of the average number of likes received by each streamer per historical live session prior to the focal session (*LnHistStreamerPop*). Compared with the original measure in the main analysis (*LnStreamerExp*), this new measure can reflect a streamer's ability, which is not subject to experience on the focal platform. The analysis results show that our conclusions are still consistent, as shown in Supplemental Table S6.

Second, there may be cases when a customer idolizes a streamer and watch the streamer's sessions, leave comments and buy products without returns repeatedly to show the support. These behaviors are irrational and differ from normal customers' purchase and return behaviors. To allay the concerns, we compute the average ratio of repeated purchase consumers at the streamer level and present it in Figure 3. We can see that the average ratio of customers who purchase in only one session from the same streamer is 94.2%, while the ratio for purchases in two sessions drops to 3.7%. The ratios for purchases in more than three or five sessions are only 0.83% and 0.25%, respectively. This analysis indicates that there are few customers who stick to the same streamer and potentially demonstrate irrational purchase behaviors.

To further tease out the confounding effect of irrational purchase behaviors, we remove the sticky customers who make purchases in more than five (or three) sessions from the same streamer and run regression analysis again. As a result, 2805 orders and 17 customers (or 5056 orders and 63 customers) are removed and 16,514 (or 14,263) observations remained in our data set. The results are shown in Supplemental Tables S7 and S8 and are consistent mostly, except that the moderating effect of assortment size is no longer significant when removing sticky customers who have purchases in more than three

Table 4. Instrumental variable estimation results (2nd stage).

	Dependent Variable: OrderReturn			
	Model 1 (2SLS)	Model 2 (2SLS)	Model 3 (Probit IV)	Model 4 (Probit IV)
LnComment	0.1770***	0.2074***	0.5477***	0.6754***
	(0.0157)	(0.0524)	(0.0497)	(0.1706)
LnComment		0.0404***		0.1442***
**LnStreamerExp*		(0.0062)		(0.0198)
LnComment		−0.0003*		−0.0012**
**AssortSize*		(0.0001)		(0.0005)
LnComment		−0.0170**		−0.0580**
**LnOrderValue*		(0.0063)		(0.0208)
LnStreamerExp	0.0286***	−0.2155***	0.0846***	−0.7868***
	(0.0051)	(0.0376)	(0.0147)	(0.1197)
AssortSize	−0.0001	0.0022*	−0.0002	0.0079**
	(0.0001)	(0.0009)	(0.0003)	(0.0030)
LnOrderValue	0.0230***	0.1313***	0.0693***	0.4415***
	(0.0041)	(0.0385)	(0.0127)	(0.1276)
LnLikeNum	−0.0086***	−0.0091***	−0.0258***	−0.0295***
	(0.0022)	(0.0022)	(0.0066)	(0.0068)
LnViewNum	−0.1173***	−0.1204***	−0.3571***	−0.3892***
	(0.0108)	(0.0108)	(0.0345)	(0.0355)
LiveDuration	−0.0065**	−0.0051*	−0.0221***	−0.0196**
	(0.0022)	(0.0022)	(0.0064)	(0.0066)
CustomerNum	0.0003***	0.0003***	0.0008***	0.0009***
	(0.0001)	(0.0001)	(0.0002)	(0.0002)
FansNum	−0.0044***	−0.0052***	−0.0135***	−0.0161***
	(0.0010)	(0.0010)	(0.0030)	(0.0031)
OrderSize	−0.0647***	−0.0611***	−0.2214***	−0.2156***
	(0.0116)	(0.0118)	(0.0436)	(0.0449)
DiscountRatio	0.2234***	0.3408***	0.6728**	1.1629***
	(0.0661)	(0.0654)	(0.2109)	(0.2194)
LnHistStreamerPop	−0.0062+	−0.0013	−0.0184+	−0.0021
	(0.0035)	(0.0036)	(0.0101)	(0.0106)
IsPlatform	−0.0532***	−0.0619***	−0.1724***	−0.2212***
	(0.0133)	(0.0136)	(0.0414)	(0.0431)
Fixed week effect	Yes	Yes	Yes	Yes
Fixed weekday effect	Yes	Yes	Yes	Yes
Fixed shipping province effect	Yes	Yes	Yes	Yes
Fixed product category effect	Yes	Yes	Yes	Yes
Intercept	−0.1467*	−0.3870	−1.7690***	−2.5967*
	(0.0690)	(0.3204)	(0.2286)	(1.0527)
Model Statistics				
N	19,319	19,319	19,319	19,319
F Statistics	30.87***	30.60***		

Note. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Robust standard errors are in parentheses. The smaller sample sizes in this table (19,319) compared with the total sample size (19,748) are due to the null values of historical statistics for the first sessions.

sessions from the same streamer (Supplemental Table S8). As result, to be conservative, we do not claim H4 is supported in our findings.

Last but not least, the instrumental variable (*LnLastComment*) we use may be subject to the sticky customer issue discussed above. To ally the concern, we run the regression analysis after excluding orders from consecutive purchase customers. That is, we exclude orders placed by the same customer in two consecutive sessions from the same streamer, and these customers' orders placed in all other sessions from the same streamer. After removing those consecutive purchase customers, our instrumental variable will not be subject to the sticky customer issues. The results are shown in Supplemental Table S9, and mostly robust, except that the moderating effect of assortment size is no longer significant.

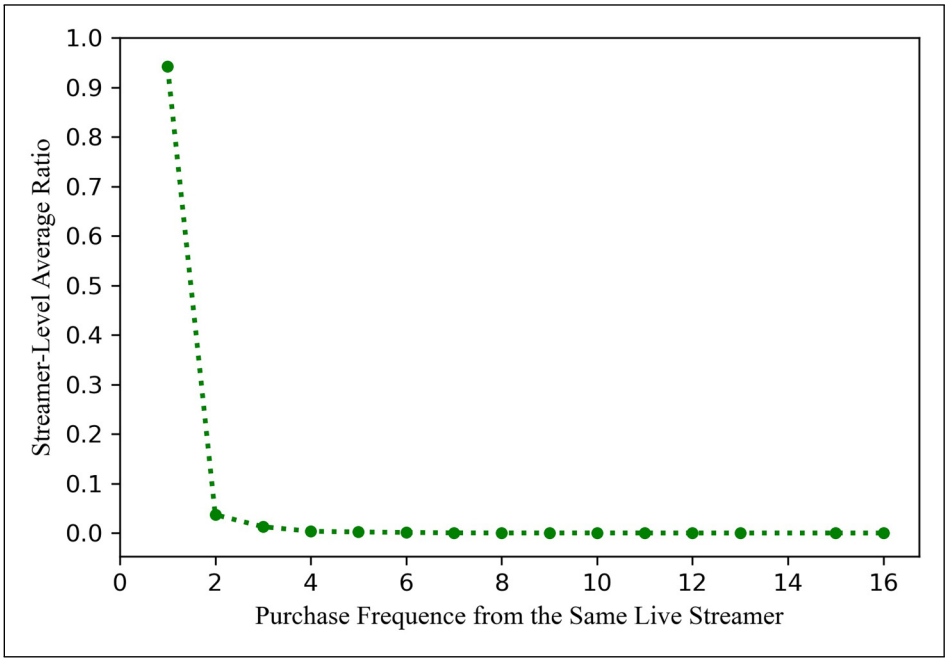


Figure 3. Customer purchase frequency from the same live streamer.

5.7 Magnitude of Persuasion Effect

To show that our results are also economically significant, we calculate the changes in return rates due to the changes of the independent variable, as well as calculate the effect size change with the changes in moderators, based on the estimation results of Model 2 in Table 4. First, we increase the number of comments by one standard deviation from its mean value and maintain the other three moderator variables at their mean levels and find that the resulting return probability increases by 11.68 percentage points in terms of absolute product return rates, representing an average increase rate of more than 30% of the average return probability in our data sample. We then draw the moderation effect plot for each of the two significant moderators, respectively. The results are presented in Figure 4. The left part of Figure 4 shows that the effect size of the natural logarithm of the number of comments on product return rates increases [decreases] from 0.17 to 0.22 [0.13] when we increase [decrease] the natural logarithm of the streamer experience by one standard deviation from its mean value and maintain the other variables at their mean levels. In contrast, the right part of Figure 4 indicates the scenario where we change the natural logarithm of the order value from its mean to its mean plus [minus] one standard deviation, leading to that the effect size changing from 0.17 to 0.15 [0.19]. Thus, the streamer experience strengthens the persuasion effect, whereas the product price weakens the persuasion effect.

We contrast the effect size of live social interaction on product return rates in our study with several studies that empirically investigate other determinants of product return likelihood in various contexts. The comparison results are presented in Table 5.

6 Conclusion

Livestream e-commerce has become increasingly popular among retailers with projected high growth in the next few years. Although prior studies have demonstrated the livestream e-commerce increase firms' sales, few studies have examined the effect on product returns. Without understanding the effect on product returns, our understanding of the holistic benefits of livestream e-commerce is limited because the increased sales may be offset by increased product returns, resulting smaller benefits than expected. The theoretical underpinning for the increased sales is the persuasion effect from the live social interactions. However, such persuasion effect may also increase the expectation gaps between a customer's perception and actual experience of a product's quality, thereby leading to greater product returns. To fill the literature gaps, we collected unique data from a leading livestream e-commerce firm and develop econometric models to study the product returns and the moderating factors that may drive the product returns.

To isolate the effect of live social interactions, we compare the product returns between live with rerun sessions. The differences between live and rerun sessions offer us a highly controlled environment in which the only difference between them is that live session has live social interactions and return session does not. We use the PSM method to match orders and conduct tests between matched orders. Furthermore, using only live orders, we develop econometric models to estimate the effect of live social interaction intensity on product returns.

We find that product returns are greater for the live sessions than those for the rerun sessions, and that the magnitude of

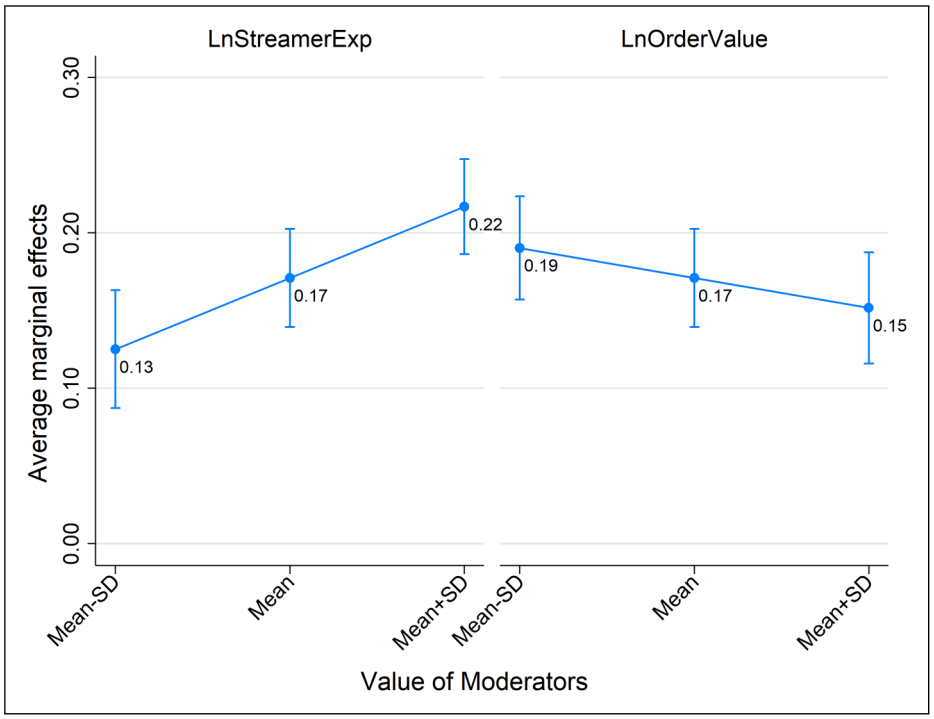


Figure 4. Moderation effect plot for the magnitude of persuasion effect.
Note: Calculations are based on estimates of Model 2 in Table 4 by fixing other variables at the mean values.

the effect increases with live social interaction intensity, confirming the existence of the persuasion effect in livestream e-commerce. The persuasion effect from live social interaction amplifies customers' prepurchase expectations but does not affect their post purchase experiences, which exaggerates the expectation gaps between prepurchase and post purpose experience and thus triggers customer dissatisfaction, resulting in higher product returns. We find the magnitude of the persuasion effect is also significant. One standard deviation increase in live social interaction intensity can increase the product return rates by as much as 11.68% in terms of absolute product return rates, which is more than 30% of the average product return rates. Our findings supplement prior studies on this subject with different findings. The prior studies in the literature have demonstrated that live social interactions are beneficial in increasing sales conversions (e.g. Song et al., 2021), and new technologies in e-commerce are beneficial in reducing product returns (e.g. De et al., 2013; Gallino and Moreno, 2018). Our findings provide a different perspective that the live social interactions and new forms of e-commerce technologies and models may be harmful in increasing the product returns.

We also find that the persuasion effect of live social interactions on product returns is stronger when the streamer has more experience on hosting livestream sessions and the product price is lower. The findings demonstrate several mechanisms through which the persuasion effect works. When the streamer gains more experience in hosting stream sessions, the streamer becomes more persuasive in pushing the sales, resulting in larger perception gaps between the expected and actual performance for customers. The larger gaps lead to greater number of product returns. When product price is low, the product is normally low involvement product therefore is easier to be persuaded by peripheral cues to purchase low-involvement products than high-involvement products. A low-involvement product usually has a low price and requires little to no exploration; in this case, it is easier for the seller to persuade customers to make a purchase by highlighting the product benefits or leveraging other peripheral cues. Our research complements the studies in the literature on persuasion effect in e-commerce by examining underlying mechanisms through which the persuasion effect works in livestream e-commerce such as source expertise (Karmarkar and Tormala, 2010; Wilson and Sherrell, 1993), source credibility (Jain and Posavac, 2001; Tybout, 1978) and product involvement (Johnson and Eagly, 1989; Sun et al., 2021).

Our results have several important managerial implications. First, for livestreaming e-commerce firms, they should be aware of the adverse persuasion effect on product returns and help prevent sessions from using excessive persuasion, which causes spikes in product returns. Livestream e-commerce managers should understand that increasing sales volume should not be the sole objective, as decreasing product return is equally important. To enhance net profits and promote sustainable growth in the long run, the managers should use both the sales volume and return rate as performance metrics for streamers. By including both metrics, the streamers would be

Table 5. Comparing the effect sizes of determinants of product returns.

Category	Study	Context	Determinants	Effect Size on Product Returns
Technology	De et al. (2013)	Online retail, a retailer that sells women's clothing products	Product-oriented web technologies: a) Zoom usage b) Alternative photo usage c) Color swatch	a) *One unit increase in zoom usage is associated with a 7% decrease in the odds for returning a product* (its odds ratio is 0.93) b) *One unit increase in alternative photo usage, is associated with increasing the odds for returning a product by 5% (its odds ratio is 1.05)* c) Color swatch technology has an insignificant effect on returns
	Gallino and Moreno (2018)	Online retail, a retailer that sells apparel in Latin America	Virtual fitting room technology	*The marginal effect of Metail* (the virtual fitting room technology) *on the probability of returning an order is −0.12 percentage points; it changes the probability of returning from 2.70% to 2.58%*
Information	Sahoo et al. (2018)	Online retail, a three-brand retailer that provides apparel, accessories, decorative home items, and home furniture in North America	Online product reviews: a) Number of reviews b) Average ratings c) Number of "helpful" reviews d) Number of reviews from "top reviewers"	a) *For 10 additional available reviews on a product, the probability of product return reduces by 1.01% (relative to the base probability of product return)* b) *0.1 higher average ratings leads to 0.94% lower probability of return* c) *A 10% increase in "helpful" reviews is associated with a 0.44% decrease in probability of return* d) *A 10% increase in reviews from "top reviewers" is associated with a 0.24% increase in the probability of return*
	Wang et al. (2021)	Online retail, a specialty retailer of outdoor goods in the western United States	Fit information in online product reviews	*The availability of treatment reviews* (i.e. fit valence and fit reference) *can reduce the return rate by 5.3%∼5.4%*
	Calvo et al. (2023)	Online retail, a leading flash-sales retailer	Information-disclosure policy	*The disclosure of low product availability increases product return rates by 17.0%*
Channel	Bell et al. (2018)	Omnichannel retail, an online-first retailer in the United States	Offline showrooms	*The average total marginal effect of showrooms on returns is −0.011, meaning that once a showroom opens, returns fall by 1.1%*
	Narang and Shankar (2019)	Omnichannel retail, a large U.S.-based retailer of video games, consumer electronics, and wireless service	Mobile app introduction	*App adopters return 35% more frequently than non-adopters in the period after app introduction*
	Our study	Livestream e-commerce, a multi-category and multi-brand platform in China	Live social interaction: number of live comments	One standard deviation increase of the number of live comments increases the return probability by 11.68 percentage points (more than 30% of the average return probability)

properly incentivized to exert an appropriate degree of persuasion in calling on consumers to make more rational purchases and thus resulting in fewer product returns. Second, since the persuasion effects are stronger when experienced streamers or lower-price products are involved, livestream e-commerce managers should think about ways to facilitate or hinder the excessive persuasion effects in different settings. For example, they should pay more attention when they use an experienced streamer or sell relatively inexpensive products. It is therefore of great interest for the livestream e-commerce firms to adopt a more holistic performance evaluation system to better incentivize streamers.

In conclusion, our research takes a first step toward studying the adverse effect of live social interactions in livestream e-commerce through the analysis of product returns. Our research findings can provide a direct reference with respect to the estimated economic loss from the increase in live social interaction intensity for livestream retailers who employ their own streamers—although not for retailers who contract with celebrities or top influencers as streamers to sell products through external livestream sessions, as our research data correspond only to the former type of streamers. Nonetheless, there are reasons to believe that the economic loss of the latter could be greater, since the higher attractiveness of celebrities or top influencers in comparison to ordinary streamers enhances the persuasion effect (Petty and Cacioppo, 1986). Thus, our research provides a lower bound on the impacts of live social interaction on product returns in the overall livestream e-commerce context.

Due to the unavailability of data, our research is not able to incorporate factors such as product portfolio in a session, information of comment contents and streamer characteristics in more detail. For instance, assortment diversity in the product portfolio may weaken the persuasion effect, as it is more difficult for the streamer to persuade customers when the product assortment is more diverse given that introducing the information about these products is more time and effort-consuming. The content of comments may also matter in the information effect, since some comments may trigger more information to be provided by the streamer than others. The demographics and cross-platform followers of streamers may contribute to the streamer heterogeneity in the persuasion effect. Future research can overcome these limitations of our research by employing richer data. Although the sales inflation is not a concern in our research context wherein the firm is a publicly listed company, future research should be careful to address the issue of sales inflation in livestream e-commerce due to that it can lead to biased result in the empirical analysis. Future research can study the impact of streamer action on product returns more deeply through video analytics, from which the researcher can better capture the persuasive attributes of streamers. Future research can also build on our findings to study customers' behaviors in livestream e-commerce by using laboratory or field experiments to address this important issue.

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Notes

1. To demonstrate robustness, we also remove orders placed within 60 min of the beginning of the rerun session, and obtain consistent results, as shown in Empirical context section. In particular, given the total number of rerun orders in the sample is 876, the 60-min cutoff removes 615 orders (with 261 orders left), and then the 24-h cutoff removes 76 orders (with 185 orders remaining).
2. Note that we do not consider the session- or streamer-related variables in this first-step propensity score equation, as we implement exact matching on orders from the same live and rerun sessions in the second step.
3. To demonstrate robustness, we use 60 min after the livestream end time as a cutoff time to grab rerun orders, and obtain consistent results, as shown in Supplemental Table S3.
4. For the variables that are highly skewed, we follow the common practice to use a natural logarithm transformation for the variables (e.g. Sun et al., 2021; Zeng et al., 2020; Zhao et al., 2021).
5. This variable shows the number of customers in thousands a streamer has on the focal platform. The streamers in our setting are sales representatives of the stores (i.e. store-run streamers), not top influencers with a large number of followers across different platforms. Though we do not have any data on other platforms, the store-run streamers likely have inconsequential numbers of followers/fans on Twitter or Facebook.
6. https://www.stata.com/support/faqs/statistics/durbin-wu-hausman-test/
7. However, as shown later in the robustness checks, the moderating effect of *AssortSize* is insignificant. As a result, we do not claim H4 is supported in our findings.

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